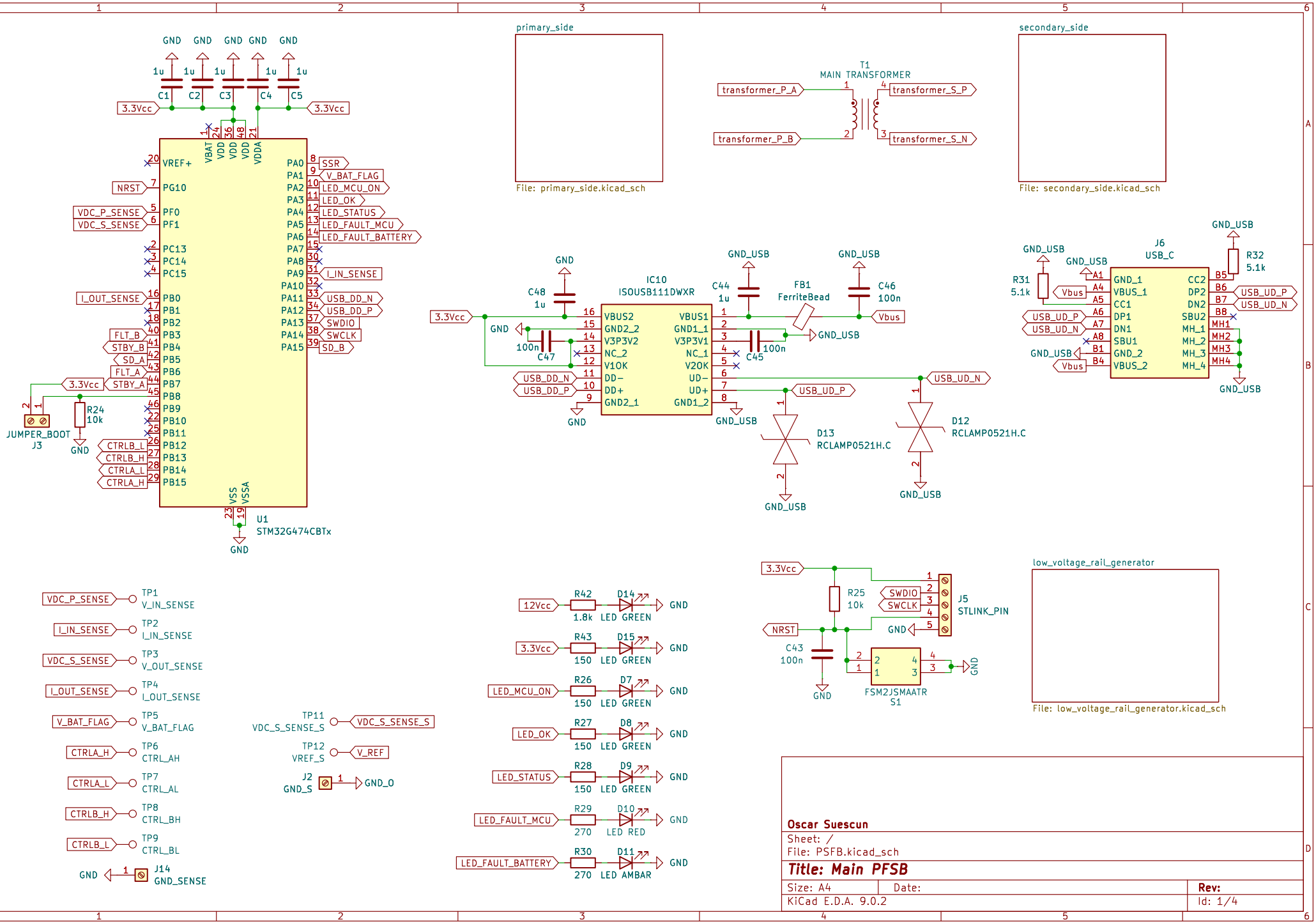
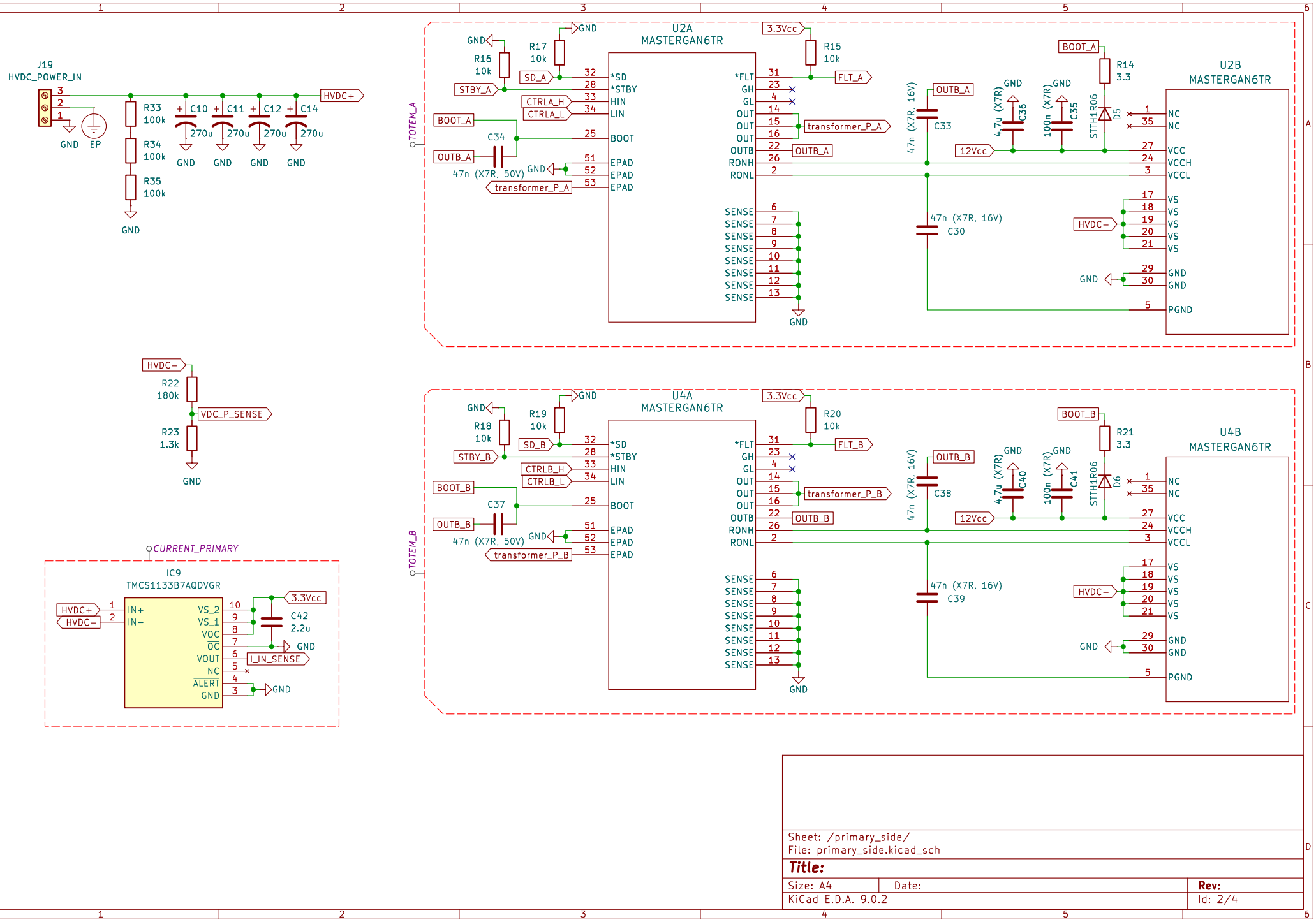
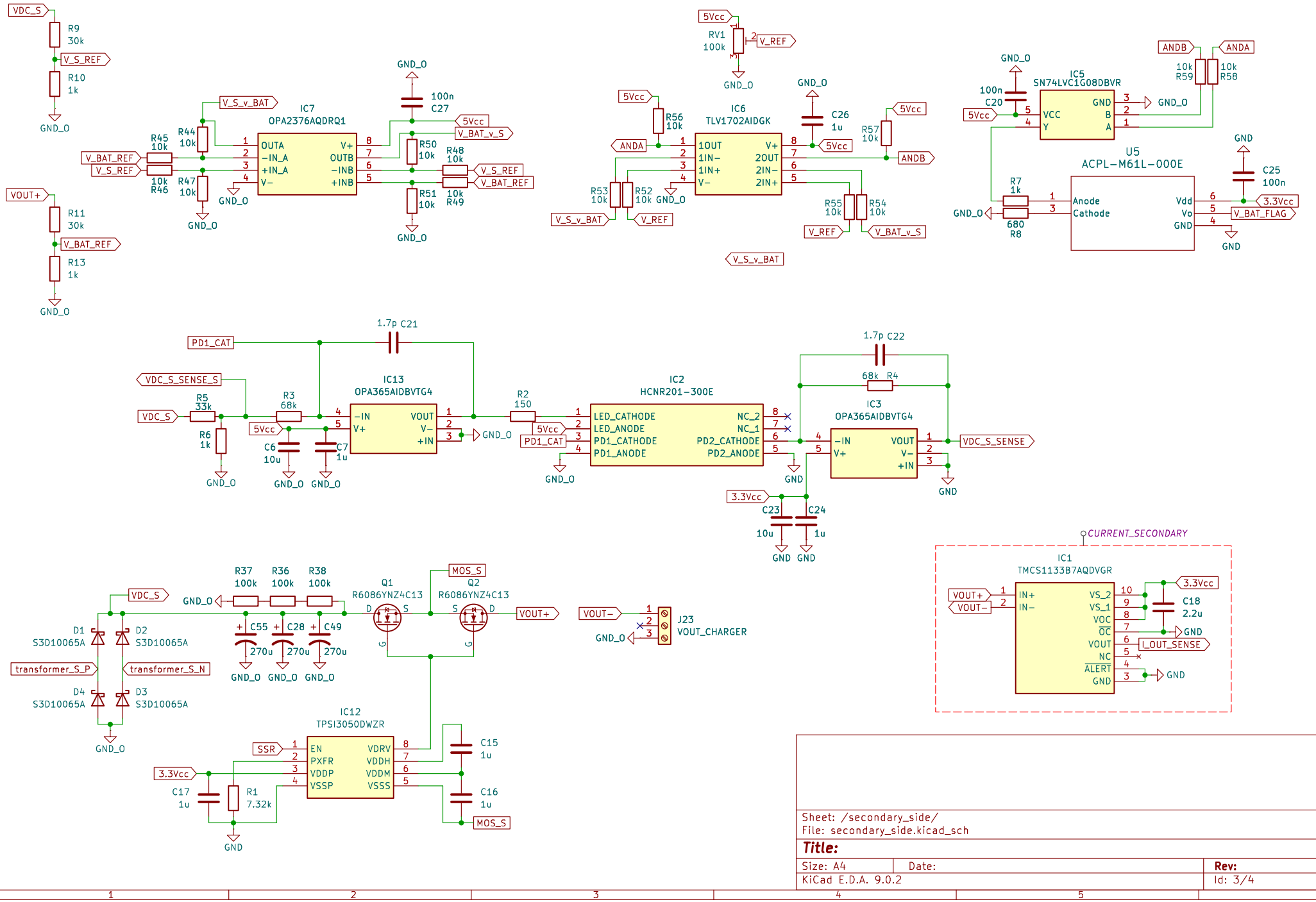
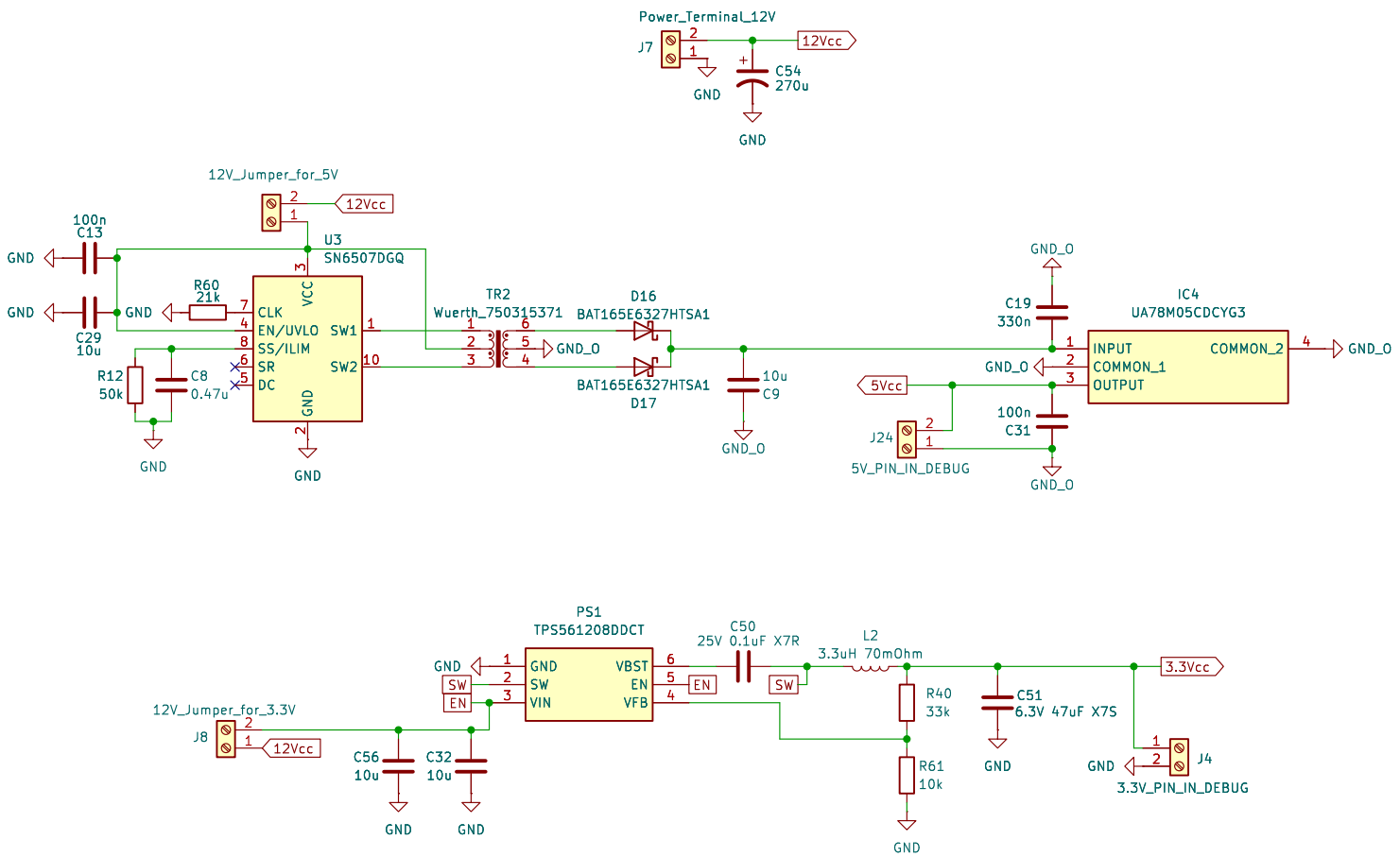

GND GND GND GND GND
1u 1u 1u 1u 1u
C1 C2 C3 C4 C5
3.3Vcc
VBAT VDD VDD VDD VDDA
VREF+ NRST VDC_P_SENSE VDC_S_SENSE I_OUT_SENSE FLT_B STBY_B SD_A FLT_A STBY_A
PG10 PF0 PF1 PC13 PC14 PC15 PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
SSR V_BAT_FLAG LED_MCU_ON LED_OK LED_STATUS LED_FAULT_MCU LED_FAULT_BATTERY I_IN_SENSE USB_DD_N USB_DD_P SWDIO SWCLK SD_B
VSS VSSA
U1
STM32G474CBTx
JUMPER_BOOT
J3
R24
10k
CTRLB_L CTRLB_H CTRLA_L CTRLA_H
primary_side
File: primary_side.kicad_sch
T1
MAIN TRANSFORMER
transformer_P_A transformer_P_B transformer_S_P transformer_S_N
secondary_side
File: secondary_side.kicad_sch
IC10
ISOUSB111DWXR
VBUS2 GND2_2 V3P3V2 NC_2 V1OK DD- DD+ GND2_1
VBUS1 GND1_1 V3P3V1 NC_1 V2OK UD- UD+ GND1_2
C48 1u
C47 100n
C44 1u
C45 100n
FB1
FerriteBead
C46 100n
USB_DD_N USB_DD_P
USB_UD_N USB_UD_P
D13
RCLAMP0521H.C
D12
RCLAMP0521H.C
J6
USB_C
GND_1 VBUS_1 CC1 DP1 DN1 SBU1 GND_2 VBUS_2
CC2 DP2 DN2 SBU2 MH_1 MH_2 MH_3 MH_4
R31 5.1k
R32 5.1k
TP1 V_IN_SENSE
TP2 I_IN_SENSE
TP3 V_OUT_SENSE
TP4 I_OUT_SENSE
TP5 V_BAT_FLAG
TP6 CTRL_AH
TP7 CTRL_AL
TP8 CTRL_BH
TP9 CTRL_BL
J14 GND_SENSE
TP11 VDC_S_SENSE_S
TP12 VREF_S
J2 GND_S
R42 1.8k D14 LED GREEN
R43 150 D15 LED GREEN
R26 150 D7 LED GREEN
R27 150 D8 LED GREEN
R28 150 D9 LED GREEN
R29 270 D10 LED RED
R30 270 D11 LED AMBAR
R25 10k
C43 100n
J5
STLINK_PIN
FSM2JSMAATR
S1
low_voltage_rail_generator
File: low_voltage_rail_generator.kicad_sch
Oscar Suescun
Sheet: /
File: PSFB.kicad_sch
Title: Main PFSB
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File: primary_side.kicad_sch		
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VDC_S
R9
30k
V_S_REF
R10
1k
GND_O
VOUT+
R11
30k
V_BAT_REF
R13
1k
GND_O
V_S_v_BAT
IC7
OPA2376AQDRQ1
R44
10k
R45
10k
V_BAT_REF
V_S_REF
R46
10k
R47
10k
OUTA
-IN_A
+IN_A
V-
V+
OUTB
-INB
+INB
GND_O
100n
C27
5Vcc
V_BAT_v_S
R50
10k
R48
10k
V_S_REF
V_BAT_REF
R49
10k
R51
10k
GND_O
5Vcc
RV1
100k
V_REF
GND_O
5Vcc
R56
10k
ANDA
IC6
TLV1702AIDGK
R53
10k
R52
10k
GND_O
V_S_v_BAT
V_REF
1OUT
1IN-
1IN+
V-
V+
2OUT
2IN-
2IN+
GND_O
C26
1u
5Vcc
R57
10k
ANDB
R55
10k
R54
10k
V_REF
V_BAT_v_S
V_S_v_BAT
GND_O
100n
C20
5Vcc
IC5
SN74LVC1G08DBVR
VCC
Y
GND
B
A
GND_O
ANDB
ANDA
10k
R59
10k
R58
U5
ACPL-M61L-000E
R7
1k
680
R8
GND_O
Anode
Cathode
Vdd
Vo
GND
GND
C25
100n
3.3Vcc
V_BAT_FLAG
GND
PD1_CAT
1.7p
C21
VDC_S_SENSE_S
VDC_S
R5
33k
R6
1k
GND_O
R3
68k
5Vcc
C6
10u
C7
1u
GND_O
GND_O
IC13
OPA365AIDBVTG4
-IN
V+
VOUT
V-
+IN
GND_O
R2
150
5Vcc
PD1_CAT
GND_O
IC2
HCNR201-300E
LED_CATHODE
LED_ANODE
PD1_CATHODE
PD1_ANODE
NC_2
NC_1
PD2_CATHODE
PD2_ANODE
1.7p
C22
68k
R4
IC3
OPA365AIDBVTG4
-IN
V+
VOUT
V-
+IN
VDC_S_SENSE
GND
GND
3.3Vcc
C23
10u
C24
1u
GND
GND
CURRENT_SECONDARY
IC1
TMCS1133B7AQDVGR
VOUT+
VOUT-
IN+
IN-
VS_2
VS_1
VOC
OC
VOUT
NC
ALERT
GND
3.3Vcc
C18
2.2u
GND
I_OUT_SENSE
GND
R37
100k
R36
100k
R38
100k
GND_O
VDC_S
Q1
R6086YNZ4C13
Q2
R6086YNZ4C13
MOS_S
VOUT+
D1
S3D10065A
D2
S3D10065A
D4
S3D10065A
D3
S3D10065A
transformer_S_P
transformer_S_N
GND_O
C55
270u
C28
270u
C49
270u
GND_O
GND_O
GND_O
VOUT-
GND_O
J23
VOUT_CHARGER
IC12
TPSI3050DWZR
SSR
EN
PXFR
VDDP
VSSP
VDRV
VDDH
VDDM
VSSS
3.3Vcc
C17
1u
R1
7.32k
GND
C15
1u
C16
1u
MOS_S
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Title:		
Size: A4	Date:	**Rev:**
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